25

INTERVIEW
QUESTIONS + ANSWERS

FIVE CONTROLS

01

EVIDENCE THRESHOLDS

Answer only when the support score clears a cost-based bar

02

ABSTENTION

Say "I do not know" and route the case, on purpose

03

CITATION VERIFICATION

Every claim maps to a passage that entails it

04

RETRIEVAL FAILURES

Find the miss, the stale page and the conflict

05

CONSTRAINED GENERATION

Schemas, grammars and quotes that limit the output

OVERVIEW
ROADMAP
25 INTERVIEW Q&A
CHEAT SHEET

GROUNDED OR SILENT

Hallucination *Reduction*

Evidence thresholds, abstention, citation verification, retrieval failures and constrained generation

An overview, a learning roadmap, 25 interview answers with numbers and a one-page cheat sheet.

1 RETRIEVED EVIDENCE

retrieval threshold τ_r = 0.45

doc_17	0.82	"return within 30 days"
doc_04	0.77	"15% restocking fee"
doc_51	0.31	"gift cards cannot be..."

BELOW τ_r: NOT IN CONTEXT

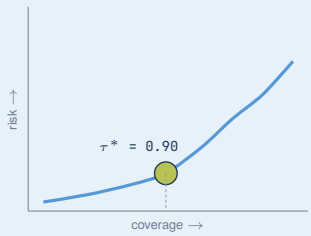
2 DRAFT ANSWER, CHECKED CLAIM BY CLAIM

- Returns are accepted within 30 days. — [1] ENTAILED 0.94
- Opened electronics carry a 15% fee. — [2] ENTAILED 0.88
- ~~Gift cards can be refunded for cash.~~ — NO SUPPORT → CUT

3 SHIPPED ANSWER

Returns are accepted within 30 days [1]. Opened electronics carry a 15% fee [2]. *I cannot confirm the gift card rule from these sources. I have sent it to a specialist.*

RISK VS COVERAGE




```
answer if support(claim, evidence) >= tau for every claim
else cut the claim, retrieve again, or abstain
```

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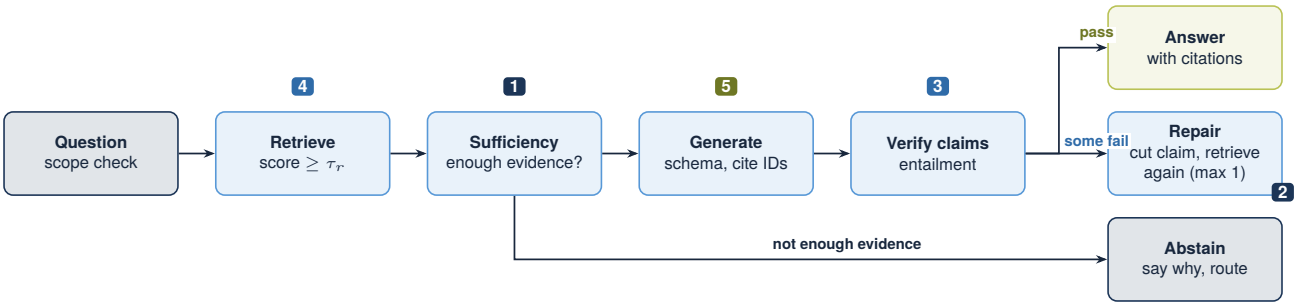
PART I

Overview

a hallucination is an unsupported claim

A hallucination is a claim in the output that the evidence does not support. Two types are important in production. A **faithfulness** error contradicts the given context or adds facts that are not in it. A **factuality** error contradicts the real world. In a retrieval-augmented (RAG) system, you control the context, so you can measure faithfulness claim by claim. Factuality then depends on two things: the context is correct, and the model stays faithful to it.

Models hallucinate for known reasons. Pre-training teaches the model to continue text, not to know where its knowledge ends. Evaluations that give zero points for "I do not know" reward guessing (Kalai et al., 2025). Retrieval can return the wrong passage, an old passage or nothing useful. Free-form decoding lets the model write any value, ID or number. No single fix removes all of these. A production system uses five controls in sequence.



EVERY ANSWER, CUT AND ABSTENTION IS LOGGED · LABELS RECALIBRATE THE THRESHOLDS

Control	Purpose	Main metric	Typical failure
1 Evidence thresholds	Answer only when the support score is high enough	Hallucination rate at a given coverage	Scores not calibrated; one global threshold
2 Abstention	Decline on purpose, say why and route the case	Abstain precision; over-abstention on answerable cases	Model refuses too much, or never
3 Citation verification	Prove that each claim comes from a cited passage	Citation precision and recall	Citation looks correct but does not entail the claim
4 Retrieval failures	Detect a missed, stale or conflicting context	Recall@k; sufficient-context rate	Model fills the gap from its own memory
5 Constrained generation	Limit the output to valid forms and values	Schema-valid rate; invented-value rate	Valid JSON that holds an invented value

Where hallucinations come from

Source	Example	Main control
No evidence in the corpus	"What is the refund rule for gift cards?" and no page covers it	Sufficiency check, abstention
Retrieval miss	The correct page exists but ranks 14th	Hybrid search, reranker, recall tests
Stale or conflicting evidence	Two policy versions disagree	Effective dates, authority rank, show the conflict
Unfaithful generation	Context says 15%, answer says 25%	Claim verification, number checks
Free-form values	The model invents an order ID or a product code	Constrained decoding, lookup validation
Pressure to answer	Prompt or metric punishes "I do not know"	Abstention in the prompt, the data and the score

KEY INSIGHT You cannot remove hallucination from a language model. You can control where it reaches the user. Make every claim pass a gate: evidence exists, the evidence entails the claim, and the form of the output is valid. When a claim fails the gate, cut it, retrieve again or abstain. Measure the result as a curve (hallucination rate against coverage), not as one number.

PART II

Learning Roadmap

from plain RAG to measured, calibrated grounding

Learn measurement before mitigation. Each level ends with a thing that you build and measure. Use one question set for all levels, for example a support-policy assistant with 300 labeled questions, so that you can compare each change with the last one.

Level	What to learn	Why it matters	After this, you can build or explain
1. Measure 1 week	Faithfulness and factuality; atomic claims (FActScore); NLI entailment; LLM-as-judge with rubrics; human labels; answerable and unanswerable sets.	You cannot reduce what you do not measure claim by claim.	A labeled set of 300 questions with a claim-level hallucination rate.
2. Retrieval quality 2 weeks	Hybrid search (BM25 + dense); rerankers; chunking; metadata and dates; recall@k; failure taxonomy; lost in the middle; query rewriting.	Most RAG hallucinations start with bad or missing context.	A retrieval test set with recall@k per failure type.
3. Thresholds and abstention 2 weeks	Calibration (reliability diagrams, ECE, temperature scaling); selective prediction; risk-coverage curves; cost-based thresholds; sufficiency classifiers; abstention messages.	The threshold sets the error rate and the coverage at the same time.	A threshold from costs, with a risk-coverage curve and an over-abstention rate.
4. Citation verification 2 weeks	Citation precision and recall (ALCE); attribution (AIS); claim decomposition; NLI verifiers; number, entity and negation checks; post-hoc repair.	A citation that looks correct is not a proof.	A verifier with its own measured precision and recall on labeled pairs.
5. Constrained generation and operations 3 weeks, then ongoing	JSON Schema and structured outputs; grammar-constrained decoding (Outlines, GBNF); enums from retrieved data; extractive quotes; regression gates; online monitoring; user feedback.	Senior interviews test the full loop: design, measure, ship, improve.	A release gate per failure slice and a dashboard with alerts.



How interviewers test this

Hallucination questions appear in RAG design, LLM evaluation and applied ML rounds. A **design round** gives a grounded assistant ("a support bot that answers from policy pages", "a clinical summary tool") and asks how you stop invented facts. A **numbers round** asks you to set a threshold and read a risk-coverage curve. A **debug round** shows a wrong answer with a correct-looking citation and asks you to find the cause. In all three, the interviewer listens for two things: do you work at the claim level, and do you measure both errors and abstentions?

Strong and weak signals

Strong signal	Weak signal
Splits the answer into atomic claims	Scores the whole answer once
Sets the threshold from error and abstain costs	"We use 0.7 similarity"
Verifies that the cited passage entails the claim	"The model adds citations"
Diagnoses retrieval before the prompt	"Tell the model not to hallucinate"
Reports a risk-coverage curve	Reports one accuracy number

RULE OF THUMB Track two numbers together: the hallucination rate on answered questions and the over-abstention rate on answerable questions. A change that improves one and hides the other is not an improvement. Always show both in the release report.

PART III

Design Building Blocks

thresholds, claims, citations, retrieval, constraints

A Cost-based evidence threshold

Answer when the expected cost of a wrong answer is smaller than the cost of an abstention. With a calibrated support score s, the error probability is $1 - s$:

$$\text{answer if } (1-s)\cdot C_{\text{wrong}} \le C_{\text{abstain}} \;\Rightarrow\; \tau^* = 1 - \frac{C_{\text{abstain}}}{C_{\text{wrong}}}$$

Example: a wrong policy answer costs $50 on average (a reopened ticket, a goodwill credit). An abstention costs $5 (a hand-off to a person). Then $\tau^* = 1 - 5/50 = 0.90$. The formula is valid only if s is calibrated. Check it with a reliability diagram on labeled cases. Set a separate τ for each risk tier.

B Claim-level verification

1. Split the draft into atomic claims (one fact each).
2. Map each claim to its cited passage IDs.
3. Score entailment: passage → claim (entailed, neutral, contradicted).
4. Run exact checks: numbers, dates, names, negation.
5. Keep entailed claims. Cut or repair the others.
6. If a required claim is cut, retrieve again once, then abstain.

C Citation metrics

Metric	Definition
Citation recall	Share of claims that their citations fully support
Citation precision	Share of citations that are needed to support their claim
Claim support	Share of atomic claims entailed by any retrieved passage
Unsupported rate	Claims with no entailing passage / all claims

Recall finds unsupported claims. Precision finds padding: citations that do not help.

D Retrieval failure taxonomy

Failure	Detect	Fix
No answer in corpus	Sufficiency check fails	Abstain; log a content gap
Missed passage	Recall@k on labeled set	Hybrid search, reranker, rewrite
Wrong chunk	Answer span cut at a boundary	Overlap, parent chunk
Stale	Effective date before change	Date filter, freshness rank
Conflict	Two passages disagree	Authority rank; show both
Distractor	Similar text, other product	Metadata filter, rerank
Buried	Key passage in the middle	Fewer chunks, reorder

E Constraint ladder

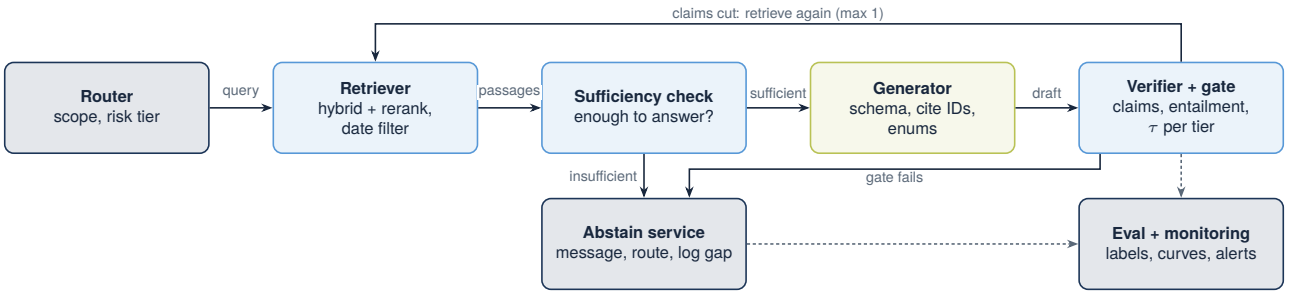
Use the strongest form that the task allows: **1** free text with citation IDs; **2** JSON Schema (structured outputs); **3** enums built from the retrieved records; **4** grammar-constrained decoding; **5** extractive span or exact quote; **6** template filled from a database lookup. A higher rung gives the model fewer ways to invent a value.

POLICY GROUNDING_POLICY.YAML

```yaml
retrieval:
  top_k: 8
  min_score: 0.45          # tau_r, after rerank
  max_retries: 1
answer_gate:
  tau: 0.90                # 1 - 5/50
  tau_by_tier: {low: 0.80, high: 0.95}
  require_citation: every_claim
verifier:
  model: nli-xl@2026-09
  exact_checks: [numbers, dates, negation]
abstain:
  message: partial_plus_reason
  route_to: support_queue
```

Reference architecture

The retriever finds evidence. The generator writes only from that evidence. The verifier and the gate decide what the user sees. The model does not decide by itself that it is grounded.



Component	Responsibility	Behavior when it fails
Retriever	Top-k passages with scores, dates and source authority	Return "no evidence"; never return an empty context silently
Sufficiency check	Decide if the passages can answer the question	Fail closed: route to abstain
Generator	Write only from the context; cite passage IDs; follow the schema	Reject invalid output; retry once with the error
Verifier + gate	Score each claim; apply τ; cut, repair or pass	Fail closed: abstain, never ship unverified claims
Eval + monitoring	Sample answers, label claims, update curves and τ	Keep the last approved thresholds

Worked example: the threshold, the curve and the cost

The script `grounding_sim.py` in this folder simulates 10,000 questions per day. 85% of the questions have an answer in the corpus. The support score is calibrated. A wrong answer costs $50 and an abstention costs $5. The formula gives $\tau^* = 0.90$. The simulation agrees: the cost is lowest at 0.90.

τ	Coverage	Abstain per day	Wrong answers per day	Rate on answered	Daily cost
Always	100%	0	1,760	17.6%	$88,019
0.70	84.1%	1,587	636	7.6%	$39,714
0.80	79.7%	2,030	531	6.7%	$36,715
0.90	**60.5%**	**3,951**	**266**	**4.4%**	**$33,032**
0.95	36.0%	6,404	88	2.4%	$36,413
0.98	15.0%	8,503	16	1.1%	$43,307

Read the table. A threshold moves along one curve. It does not change the curve. At 0.90, the system answers 60.5% of questions and 4.4% of those answers are wrong. If policy needs a rate below 2% with more than 60% coverage, no threshold can give it. Improve the system instead: better retrieval and a better verifier push the curve down.

Test the verifier too. The same script runs a lexical citation check on 8 labeled pairs. It catches a wrong number (15% vs 25%). It misses a negation ("can be refunded" vs "cannot be refunded") and rejects a correct paraphrase. Precision and recall are both 0.75. Measure your verifier the same way before you trust it.

IV

25 INTERVIEW QUESTIONS · Q01 TO Q05

Evidence Thresholds

when is the evidence strong enough to answer?

Q01 What is a hallucination, and how do you measure the rate? CORE

A hallucination is a claim that the evidence does not support. Measure it at the claim level, not the answer level. Split each answer into atomic claims. Label each claim as supported, contradicted or not verifiable against the retrieved context (faithfulness) or a trusted source (factuality). FActScore uses this method (Min et al., 2023). Report two rates: unsupported claims per claim, and answers with at least one unsupported claim. Use an answerable set and an unanswerable set. **TEST** Two people label 100 answers. Agreement (Cohen's κ) must be above 0.7 before you use the labels for a release decision.

> **INTERVIEWER LENS** Say "claim-level" and "faithfulness vs factuality" in the first minute.

Q02 How do you set the evidence threshold for answering? SENIOR

Start from costs, not from a round number. Estimate the cost of a wrong answer and the cost of an abstention. Compute $\tau^* = 1 - C_{\text{abstain}}/C_{\text{wrong}}$ (Block A). Then calibrate the score. Fit temperature scaling or isotonic regression on held-out labeled cases (Guo et al., 2017). Plot the risk-coverage curve and mark τ^* on it. Set one threshold per risk tier: a medication question and a store-hours question do not have the same error cost. **NUMBERS** $50 / $5 gives 0.90; in the worked example this answers 60.5% at a 4.4% error rate. **TEST** Recompute the reliability diagram each month on reviewed cases.

Q03 Which signals tell you that the evidence supports an answer? SENIOR

Combine signals from three places. **Retrieval**: top reranker score, score gap to the next passage, number of agreeing passages. **Verification**: NLI entailment of each claim by its passage, exact match of numbers and names. **Generation**: self-consistency across samples (Manakul et al., 2023), semantic entropy over meaning clusters (Farquhar et al., 2024), token log-probabilities. Train a small model that maps these signals to the probability that the answer is correct, with human labels. Calibrate it. Do not use the model's verbal "I am 90% sure" as the score. **TEST** Compare each signal and the combination by AUROC on labeled answers. Keep a signal only if it adds lift.

Q04 What is the difference between a retrieval threshold and an answer threshold? MID

The **retrieval threshold** τ_r decides which passages enter the context. A low τ_r adds distractors. A high τ_r drops the passage that holds the answer. The **answer threshold** τ decides if the final answer ships. It uses all signals, not only retrieval scores. Tune τ_r for recall of the gold passage at a fixed context size. Tune τ for the cost trade-off. Do not use a raw cosine similarity as the answer threshold: it measures topic overlap, not support. **TEST** For τ_r: gold-passage recall@k $\geq$ 95% on the labeled set. For τ: the risk-coverage target per tier.

Q05 How do you keep thresholds valid after a model or index change? MID

A threshold belongs to one model, one verifier and one index. A new embedding model changes the score scale. A new LLM changes the error rate at each score. After each change, rerun the labeled set, refit the calibration and recompute τ. Watch for drift in production: track the score distribution and the abstention rate per day. A sudden rise or fall of the abstention rate is a signal, not a success. **TEST** The CI gate blocks a release when the calibration error (ECE) rises by more than 50% or when coverage at the target error rate falls by more than 3 points.

25 INTERVIEW QUESTIONS · Q06 TO Q10

V Abstention

decline on purpose, say why, route the case

Q06 When should the model abstain? CORE

Abstain in five cases. **1** The corpus has no answer. **2** The retrieved evidence is not sufficient. **3** The sources conflict and policy gives no rule to choose. **4** The question is out of scope (legal advice from a shipping bot). **5** The answer is high-risk and the score is below the tier threshold. Abstention is a designed output, not a failure. The system must know the reason, so that it can log it and route the case. **TEST** An unanswerable set of 100 questions per domain. Target abstain recall $\geq$ 90% on that set.

Q07 How do you teach a model to abstain without making it refuse too much? SENIOR

Use three levers. **Prompt**: say that "not in the sources" is a valid answer and give the exact format. **Data**: fine-tune with unanswerable questions mixed in, as SQuAD 2.0 does for reading comprehension (Rajpurkar et al., 2018). R-Tuning trains the model to refuse questions outside its knowledge (Zhang et al., 2024). **Gate**: let an external threshold decide, so that the model does not carry the full load. Then measure over-abstention on an answerable set. Training on refusals alone makes the model refuse answerable questions. **TEST** Report abstain recall on unanswerable questions and over-abstention on answerable ones. Both must meet the target.

> **INTERVIEWER LENS** Strong answers name over-abstention as the second metric.

Q08 How do you evaluate abstention? SENIOR

Use selective prediction metrics. Plot **risk** (error rate on answered questions) against **coverage** (share answered) for all thresholds (Geifman and El-Yaniv, 2017). Report the area under the risk-coverage curve (AURC) and the coverage at a target risk, for example coverage at 2% risk. Add a confusion table: correct answer, wrong answer, correct abstention, wrong abstention. Score the system with the cost of each cell. Plain accuracy rewards guessing, because an abstention scores the same as a wrong answer (Kalai et al., 2025). **TEST** Release only if coverage at the target risk does not fall below the last release on any slice.

Q09 What makes a good abstention message? MID

A good abstention is useful. Say what you could confirm, with citations. Say what you could not confirm and why: no source, old source, or sources disagree. Give the next step: a link, a form, or a hand-off to a person with the context attached. Do not blame the user. Do not invent a near answer. Keep the tone the same as a normal answer. A short, specific message gets fewer repeat questions than "I cannot help with that". **TEST** Sample 50 abstentions per week. Count repeat questions within one hour. Target: below 15%.

Q10 How do you handle a partly supported answer? MID

Work at the claim level. Keep the claims that pass the gate. Cut the claims that fail. If a cut claim is necessary to answer the question, retrieve again once with a rewritten query. If it still fails, ship the supported part and state the gap: "The policy confirms X. I cannot confirm Y from the sources." Do not let the model rewrite the cut claim into a vaguer version of the same claim. Verify the final text again after repair. **TEST** Feed answers with one planted unsupported claim. The shipped answer must contain zero planted claims.

25 INTERVIEW QUESTIONS · Q11 TO Q15

VI Citation Verification

a citation is a claim that needs a check

Q11 How do you verify that a citation supports a claim? CORE

Treat each (claim, passage) pair as an entailment test. The passage is the premise. The claim is the hypothesis. A natural language inference (NLI) model or an LLM judge returns entailed, neutral or contradicted. A claim is supported only if its cited passages entail it. This follows the attributable to identified sources (AIS) idea: a reader of the source would agree that it says the claim (Rashkin et al., 2023). Add exact checks for numbers, dates and names, because NLI models often miss them. **TEST** Measure the verifier on 500 labeled pairs, with negations and swapped numbers in the set.

Q12 Define citation precision and recall. How do you measure them? SENIOR

ALCE defines them for generated text (Gao et al., 2023). **Citation recall**: a sentence scores 1 if the union of its citations entails it. **Citation precision**: a citation scores 1 if it is needed, that is, the remaining citations do not entail the sentence without it, or it supports the sentence alone. Low recall means unsupported claims. Low precision means decoration: many citations that do not help. Compute both with an NLI model, and check the NLI model against human labels on a sample. **NUMBERS** Typical release targets: citation recall $\geq$ 0.95 for high-risk tiers, precision $\geq$ 0.85.

> **INTERVIEWER LENS** Precision is the less known half. Naming it is a strong signal.

Q13 How do you stop fabricated or misplaced citations?

MID

Give each passage a short ID in the context, such as `[P3]`. Tell the model to cite only these IDs. Validate every ID after generation: an ID that is not in the context is a fabricated citation, so reject the answer. For a misplaced citation (correct document, wrong passage), verify entailment against the cited passage only. For legal or medical text, require an exact quote and check it with string match against the source. **TEST** Count citations to IDs that are not in the context. The count must be zero in every release.

Q14 Your verifier misses negations and numbers. How do you make it robust?

SENIOR

Build a test suite of hard pairs first: negation ("can" vs "cannot"), swapped numbers, swapped entities, wrong time period, partial support. Measure the verifier on each slice. Add rule checks where models are weak: extract numbers, units, dates and named entities from the claim and require them in the passage. Use an NLI model trained for factual consistency, and benchmark it (TRUE, Honovich et al., 2022). Calibrate the verifier score like any other model. Our lexical baseline in `grounding_sim.py` shows the problem: word overlap passes "gift cards can be refunded" against "gift cards cannot be refunded". **TEST** Recall on the contradiction slice ≥ 0.95 before the verifier gates releases.

Q15 How do you verify citations at low latency?

SENIOR

Verify claims in parallel, not in sequence. Use a small cross-encoder NLI model for most claims and an LLM judge only for uncertain scores. Stream the answer by sentence and hold each sentence until it passes. Cache verification for frequent (claim, passage) pairs. Verify every claim in high-risk tiers. Sample and verify offline in low-risk tiers. Set a latency budget, for example 300 ms at P95 for verification, and track it. **TEST** Load test at peak traffic. P95 total latency stays under the SLA, and no sentence ships before its verdict.

VI 25 INTERVIEW QUESTIONS · Q16 TO Q20
Retrieval Failures

find the miss, the stale page and the conflict

Q16 Which retrieval failures cause hallucinations?

CORE

Seven failures are common (Block D): no answer in the corpus, a missed passage, a wrong chunk boundary, a stale passage, conflicting passages, a distractor from a similar product, and a key passage buried in a long context. In each case, the model gets a gap and fills it from its own memory. Diagnose retrieval before you change the prompt. Check if the gold passage was in the top-k. If it was not, the problem is retrieval. If it was, the problem is generation. **TEST** Label 200 wrong answers with a failure type. Fix the largest group first.

Q17 How do you detect that the context is not sufficient?

SENIOR

Add an explicit sufficiency step between retrieval and generation. A classifier or an LLM judge reads the question and the passages and answers: can a careful reader answer from this text alone? Joren et al. (2025) show that models often answer instead of abstaining when the context is not sufficient, and that a sufficiency signal improves selective answering. Corrective RAG grades retrieved documents and triggers new retrieval when they are weak (Yan et al., 2024). Self-RAG trains the model to judge relevance and support with reflection tokens (Asai et al., 2024). **TEST** Measure the sufficiency check on labeled (question, context) pairs. Report precision and recall for "insufficient".

Q18 The sources conflict or one is stale. What does the system do?

MID

Store an effective date, a version and an authority level with each chunk. Filter by date at retrieval time when the question has a date. When two passages conflict, prefer the higher authority, then the later effective date. If policy gives no rule, show both and say that they conflict. Do not let the model merge two versions into one answer. Log each conflict as a content defect for the document owner. **TEST** Plant 20 superseded passages next to their new versions. The answer must cite the new version in all 20 cases.

Q19 Long context has the answer, but the model ignores it. Why, and what do you do?

MID

Models use information at the start and the end of a long context better than information in the middle (Liu et al., 2024). More chunks also add distractors. Retrieve more, then rerank and keep fewer: for example, retrieve 50 and pass 5 to 8. Put the strongest passages first. Remove near-duplicates. Ask the model to quote the supporting text before it answers, so that it must find it. **TEST** Move the gold passage to positions 1, 5 and 10 in the context. Accuracy must not drop by more than 3 points.

Q20 How do you build a regression set for retrieval failures?

SENIOR

Make one slice per failure type, 30 to 50 cases each. Include unanswerable questions, stale and conflicting pairs, near-duplicate products, multi-hop questions and paraphrased queries. Label the gold passages and the expected behavior: answer, partial or abstain. Run the set on every change to chunking, embeddings, reranker, prompt or model. Gate the release per slice, not on the average, because an average can hide one broken slice. **TEST** The release fails if any slice drops more than 5 points, even when the overall score rises.

VI

25 INTERVIEW QUESTIONS · Q21 TO Q25

Constrained Generation

limit what the model is able to write

Q21 How does constrained generation reduce hallucination? CORE

Constrained generation limits the output to a set of valid forms. A JSON Schema fixes the fields and types. An enum fixes the allowed values. A grammar fixes the syntax. Each constraint removes ways to invent something: a field that does not exist, a status that is not real, a malformed date. Constraints control form, not truth. A valid JSON object can still hold a wrong value. Use constraints together with verification. **TEST** Track two rates: schema-invalid outputs (target 0%) and invented values in valid outputs.

INTERVIEWER LENS "Constraints fix the form, verification fixes the content" is the key sentence.

Q22 Structured outputs or grammar-constrained decoding: what is the difference? MID

Structured outputs from an API force the response to match a JSON Schema. They are easy to use and cover most extraction tasks. **Grammar-constrained decoding** masks invalid tokens at each step with a regular expression or a context-free grammar. Outlines does this with a finite-state machine over the vocabulary (Willard and Louf, 2023). llama.cpp uses GBNF grammars. Use a grammar when you host the model or need a format that a schema cannot express. Note one risk: a tight constraint can force a low-probability token and lower quality. Leave a valid "unknown" path in every schema. **TEST** Compare task accuracy with and without the constraint on the same set.

Q23 How do you stop the model from inventing IDs, codes or entities? SENIOR

Do not let the model generate an ID from free text. Build the allowed values from the retrieved records at request time and put them in a dynamic enum or grammar. After generation, look up every ID in the system of record. Reject the output if the lookup fails. For entities, link the mention to a known entity ID and verify the link. Allow `null` with a reason, so that the model is never forced to choose a wrong value. **TEST** Invented-ID rate = IDs that fail lookup / all IDs. Target 0 in high-risk tiers.

PYTHON DYNAMIC ENUM

```python
# allowed IDs: retrieved records only
ids = [r.order_id for r in records]
schema = {"type": "object", "properties": {
  "order_id": {"enum": ids + [None]},
  "reason": {"type": "string"}},
  "required": ["order_id", "reason"]}
```

Q24 When do you quote instead of generating? MID

Quote when the exact words matter: contracts, policies, dosages, regulations. Ask the model to extract the supporting span first, then answer from the span. Verify the quote with an exact or normalized string match against the source. Show the quote with its source link. Generate freely only for the parts that need synthesis, and verify those parts as claims. **TEST** Quote match rate against the source text = 100%. A quote that does not match is a hallucination.

Q25 Design a grounded assistant for a health insurer's benefits questions. SENIOR

SCOPE Benefits and coverage only. Route medical advice and claims disputes to people. **RETRIEVE** Plan documents by plan ID and effective year; hybrid search; rerank; date filter. **GATE** Sufficiency check, then claim verification. $\tau = 0.95$ for amounts and coverage rules. **GENERATE** Schema with plan ID from an enum, amounts as quotes, citation IDs per claim. **ABSTAIN** Partial answer, reason, hand-off with the plan and the question. **MEASURE** Claim-level hallucination rate, citation recall and precision, coverage at target risk, per slice. **TEST** Swap one plan year in the context. The system must cite the correct year or abstain.

INTERVIEWER LENS Interviewers look for scope routing, plan-year filtering and quoted amounts.

Debug drill: symptom, likely cause, first check

Symptom	Likely cause	First check
Wrong answer with a correct-looking citation	Cited passage does not entail the claim	Run the verifier on that (claim, passage) pair
Confident answer to an unanswerable question	No sufficiency check; prompt punishes "I do not know"	Abstain recall on the unanswerable set
Old rule in the answer	Stale chunk outranks the new version	Effective dates and the date filter
Correct page exists, answer is still invented	Gold passage not in top-k, or buried	Gold-passage recall@k and its position
Valid JSON with an order ID that does not exist	ID generated from free text	Lookup failure rate; dynamic enum
Abstention rate doubles after a release	Score scale changed with the new model	Reliability diagram; refit calibration

RELEASE GATE

Release Checklist and References

use it before any grounded feature ships

Measurement

- ☐ Labeled answerable and unanswerable sets exist per domain.
- ☐ Hallucination rate is claim-level, with label agreement $\kappa > 0.7$.
- ☐ The report shows risk-coverage, not one accuracy number.

Retrieval

- ☐ Gold-passage recall@k is measured per failure slice.
- ☐ Chunks carry effective dates, versions and authority.
- ☐ A sufficiency check runs before generation.

Thresholds and abstention

- ☐ Scores are calibrated on recent labeled data.
- ☐ τ comes from costs, per risk tier.
- ☐ Over-abstention on answerable questions is in the report.

Citations and constraints

- ☐ Every claim cites a passage ID from the context.
- ☐ The verifier has its own precision and recall, with a negation slice.
- ☐ IDs and codes come from enums or lookups, never free text.
- ☐ Every schema has a valid "unknown" path.

Operations

- ☐ The regression gate fails on any slice drop above 5 points.
- ☐ Thresholds are recomputed after each model or index change.
- ☐ Abstention reasons feed a weekly content-gap review.

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Metrics dashboard

Metric	Definition	Example alert
Claim hallucination rate	Unsupported claims / all claims, on sampled answers	Above the tier limit for 2 days
Coverage at target risk	Share answered while risk stays at the target	Falls more than 3 points after a release
Over-abstention	Abstained / answerable, on the labeled set	Above 10% on any slice
Citation recall and precision	ALCE definitions, NLI scored, human-checked sample	Recall below 0.95 in high-risk tiers
Invented-ID rate	IDs that fail lookup / all generated IDs	Any value above 0
Abstention rate (live)	Abstained / all questions, per day	Change of more than 30% day over day
Content gaps	Top abstention reasons by topic	New topic in the top 10

PRACTICE

Mock Design Round

45 minutes, one grounded assistant, one rubric

Prompt: "Design an assistant that answers employee questions from 4,000 HR policy pages in 12 countries. Wrong answers about pay or leave are expensive. How do you keep it from inventing policy?" Use this script to practice. Speak in claims, numbers and tests.

Minute	Step	What you say	What you draw or write
0–5	Scope and risk	Users, countries, question types. Pay and leave are high-risk; office hours are low-risk. Legal advice is out of scope.	Risk tiers with τ per tier
5–10	Failure types	No policy, wrong country, old version, conflict between global and local policy, invented numbers.	Failure list = test slices
10–20	Retrieval	Hybrid search with a country and effective-date filter. Rerank 50 to 6. Authority: local overrides global.	Retriever box with metadata
20–28	Threshold and abstention	Sufficiency check. τ^* from costs: \$200 wrong, \$8 hand-off gives 0.96 for pay. Abstain with a partial answer and an HR ticket.	$\tau^* = 1 - 8/200$
28–36	Verify and constrain	Claims, cited IDs, NLI entailment, number and date checks. Country and policy ID from enums. Leave days quoted.	Verifier and gate
36–42	Measure and ship	400 labeled questions, 100 unanswerable. Risk-coverage per country. Gate per slice.	Release gate rule
42–45	Trade-offs	Coverage vs risk. Latency of verification. Content gaps go to HR owners.	Open questions

Answer gate sketch

PYTHON SKETCH

```python
def answer_gate(q, passages, draft, tier):
    if not passages:
        return "abstain", "no_evidence"
    if sufficiency(q, passages) < P.tau_suff:
        return "abstain", "insufficient_context"
    kept, cut = [], []
    for c in split_claims(draft):
        if not c.cites or not set(c.cites) <= passages.keys():
            cut.append(c); continue   # missing or invented ID
        ctx = " ".join(passages[i].text for i in c.cites)
        s = calibrated(entail(ctx, c.text))
        ok = s >= P.tau[tier] and exact_checks(c.text, ctx)
        (kept if ok else cut).append(c)
    if not kept:
        return "abstain", "no_supported_claim"
    if any(c.required for c in cut):
        return "partial", render(kept) + gap_note(cut)
    return "answer", render(kept)
```

The gate is code, not a prompt. It fails closed. It returns a reason with every abstention, so that the reason can be logged and routed.

Scoring rubric

Dimension	Hire signal	No-hire signal
Diagnosis	Lists failure types before tools	Starts with the prompt
Grounding	Claim-level verification with cited IDs	"The model adds sources"
Thresholds	Costs, calibration, τ per tier	One similarity cut-off
Abstention	Partial answer, reason, route; tracks over-abstention	Refusal as failure, or never refuses
Retrieval	Country and date filters, authority rule	Bigger top-k
Measurement	Risk-coverage per slice, release gate	One accuracy number

FOLLOW-UP QUESTIONS "Coverage fell to 40%. What now?" Find the top abstention reasons. Fix retrieval and content gaps first. Lower τ only with new cost evidence.

"Verification adds 600 ms." Verify sentences in parallel, use a small NLI model, call an LLM judge only for uncertain scores.

PART XI

Cheat Sheet

one page before the interview

Formulas

Answer threshold:
$\tau^* = 1 - C_{abs}/C_{wrong}$
Answer if: $(1 - s)\, C_{wrong} \leq C_{abs}$
Coverage: answered / all
Risk: wrong / answered
Claim support: entailed claims / all claims
Over-abstention: abstained / answerable

Two kinds

Faithfulness: output vs the given context.
Factuality: output vs the world.
RAG: measure faithfulness; fix the corpus for factuality.

Answer order (design round)

Scope → failure types → retrieval and dates → sufficiency → constrained generation → claim verification → threshold per tier → abstain path → metrics → regression gate.

Support signals

Reranker score and gap, NLI entailment, exact number and name match, self-consistency, semantic entropy, log-probs. Combine, then calibrate on labels.

Abstain when

No answer in corpus. Context not sufficient. Sources conflict. Out of scope. Score below the tier threshold.

Good abstention

What is confirmed (cited) + what is not and why + next step + hand-off with context.

Citation checks

IDs exist in context. Cited passage entails claim. Numbers, dates, negation match. Quotes match exactly. Recall and precision (ALCE).

Retrieval failures

1. No answer in corpus
2. Missed passage
3. Wrong chunk boundary
4. Stale passage
5. Conflicting passages
6. Distractor
7. Buried in the middle

Constraint ladder

Free text + IDs → JSON Schema → dynamic enums → grammar → exact quote → template from lookup.

Red flags

"Tell it not to hallucinate." Cosine 0.7 as the answer gate. Accuracy only. Citations not verified. IDs from free text. No unanswerable set.

Glossary

Term	Meaning	Term	Meaning
Atomic claim	One fact that is true or false alone	Calibration	Scores that match real accuracy
Entailment	The premise makes the claim true	Selective prediction	Answer only above a threshold
Sufficient context	Passages that can answer the question	Risk-coverage curve	Error rate vs share answered
Citation recall	Claims fully supported by their citations	AURC	Area under risk-coverage curve
Citation precision	Citations that are needed	Dynamic enum	Allowed values built per request
Over-abstention	Declining an answerable question	Semantic entropy	Disagreement in meaning across samples

Ground every claim. Make silence a feature.

Hallucination reduction is a system: retrieval that finds the right page, a threshold set from costs, citations that are checked, constraints on form and an abstain path that helps. Measure each part with its own metric. For more practitioner guides and interview prep, subscribe at aiengineeringinsider.substack.com/subscribe and find every guide at beacons.ai/aiengineeringinsider.